Geometric Acceleration on Compact S^3 Without a Fundamental Λ: A Phase–Wavefunctional Mechanism

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We present a rigorously testable framework in which late-time acceleration is generated by a *phase–geometry* mechanism on a compact, simply connected background ($k=+1$, S^3), instead of a fundamental fluid-like dark energy. In a minimal two-mode reduction of a homogeneous scalar wavefunctional the Hubble rate is tied to the relative phase drift as $H=\frac{1}{3}\dot{\Delta\theta}$. When the phase velocity approaches a constant at late times, an asymptotic effective cosmological constant $\Lambda_{\rm eff}\equiv 3H_\infty^2/c^2$ emerges. Background dynamics reduce exactly to GR with dust, small positive spatial curvature (convention $\Omega_k<0$), and the emergent $\Lambda_{\rm eff}$; no exotic fluid is introduced. We clarify that simply connected S^3 predicts no matched-circle patterns, specify a reproducible CMB+BAO+SN(+RSD) pipeline, and provide compact formulas for distances and growth on $k=+1$.

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Sign conventions. $\Omega_k \equiv -\dfrac{c^2}{a^2H^2R_{\rm curv}^2}$, so $k=+1$ (closed) $\Rightarrow \Omega_k<0$. $D_M = R_{\rm curv}\sin(\chi/R_{\rm curv})$, $\chi(z)=\int_0^z \dfrac{c\,{\rm d}z'}{H(z')}$.

I. BACKGROUND DYNAMICS AND CONSISTENCY

At late times the background is GR with dust, curvature, and an emergent constant $\rho_\Lambda=\Lambda_{\rm eff}c^2/(8\pi G)$ once $\dot{\Delta\theta}\to 3H_\infty$:

$$H^2=\frac{8\pi G}{3}(\rho_m+\rho_\Lambda)-\frac{c^2}{a^2R_{\rm curv}^2}, \qquad (1)$$

$$\dot\rho_m+3H\rho_m=0, \qquad \rho_m=\rho_{m0}a^{-3}. \qquad (2)$$

The solution approaches de Sitter with $H\to H_\infty$; the *origin* of Λ is traced to the phase drift $H=\frac{1}{3}\dot{\Delta\theta}$ (derived below and in App. A).

II. FOUNDATIONS: TOPOLOGY, CAUSALITY, ENTROPY, AND NOETHER STRUCTURE

Spatial topology. Spatial slices are $\Sigma_t\simeq S^3=\{x\in\mathbb{R}^4 : |x|=R_{\rm curv}\}$; the spacetime manifold is $\mathcal{M}\simeq\mathbb{R}\times S^3$. The slice is compact, boundaryless, and simply connected with $\pi_1(S^3)=0$. This eliminates matched-circle identifications and fixes the $k=+1$ geometry used throughout.

Universal light cone and the value of c. All fields propagate on the same metric $g_{\mu\nu}$ via minimal coupling, so the null cones coincide. The constant c is the Lorentz-invariant conversion between temporal and spatial units; in natural units $c\equiv 1$, while its SI value reflects units, not dynamics. The phase degree $\Delta\theta$ influences expansion only through the homogeneous relation $H=\frac{1}{3}\dot{\Delta\theta}$ and does not modify photon characteristics or local Lorentz symmetry.

Entropy and the arrow of time. With a low-entropy past boundary and expansion on compact S^3, coarse-grained matter entropy increases, while the horizon contribution $S_{\rm hor}\sim \pi c^3/(G\hbar H^2)$ grows as $H(t)$ decreases towards H_∞. The generalized total $S_{\rm tot}=S_{\rm matter+radiation}+S_{\rm grav}+S_{\rm hor}$ is non-decreasing, supplying a consistent arrow of time compatible with the phase-driven approach to de Sitter.

Bell nonlocality without signalling. Quantized fields on $\mathbb{R}\times S^3$ reproduce Bell-inequality violations via entanglement while preserving microcausality $[\hat{\mathcal{O}}(x),\hat{\mathcal{O}}(y)]=0$ for spacelike separations. The global homogeneous phase $\Delta\theta$ is not a signalling channel; it enters only via the background expansion.

Noether structure and conservation laws. Diffeomorphism invariance yields $\nabla_\mu T^{\mu\nu}=0$. Spatial isometries of S^3 generate the standard charges for perturbations. If $U_{\rm eff}$ is approximately flat (shift symmetry for $\Delta\theta$), the internal shift $\Delta\theta\to\Delta\theta+{\rm const}$ produces the conserved current $J^\mu_{(\theta)}=I_\theta\nabla^\mu\Delta\theta$, with homogeneous momentum $p_\theta=a^3I_\theta\dot{\Delta\theta}$. The BO friction term $3H$ implements the usual dilution of a comoving charge due to expansion and underlies the saturation $\dot{\Delta\theta}\to 3H_\infty$.

III. MICROPHYSICAL FOUNDATION: WDW → BO → PHASE LAW

Minisuperspace and Wheeler–DeWitt. For FRW with $k=+1$ and spatial S^3 (Vol $=2\pi^2a^3$), the gravity+homogeneous scalar action reduces to a 1D system in $a(t)$ and the homogeneous matter variables. Canonical

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quantization yields the WdW equation

$$\Big[-\frac{\hbar^2}{2\mathcal{M}(a)}\partial_a^2 + U_g(a)\Big]\Psi + \hat{H}_m(a,\hat{q},\hat{p})\,\Psi = 0, \quad (3)$$

with Laplace–Beltrami factor ordering and $\mathcal{M}(a) \propto a$.

Born–Oppenheimer (BO) ansatz and emergent time. Adopt

$$\Psi(a,q) = e^{\frac{i}{\hbar}S_0(a)} \sum_{i=1}^{2} A_i(a)\, e^{i\theta_i(a)}\, \psi_i(q;a), \quad (4)$$

with $\langle\psi_i|\psi_j\rangle = \delta_{ij}$ in the S^3 inner product $\propto a^3 \int \mathrm{d}^3x$. At leading WKB order, S_0 obeys the Hamilton–Jacobi equation and defines an *emergent time* through

$$\partial_t \equiv \frac{1}{a^3}\frac{\partial S_0}{\partial a}\,\partial_a, \qquad \frac{\partial S_0}{\partial a} \propto a^2 H. \quad (5)$$

At next order one obtains Schrödinger equations $i\hbar\,\partial_t\psi_i = \hat{H}_m\,\psi_i$ and transport equations for A_i, θ_i.

Two-mode reduction and effective action. Retaining two BO branches and integrating out q yields an *effective homogeneous* action for the relative phase $\Delta\theta \equiv \theta_1 - \theta_2$:

$$S_{\rm eff} = \int \mathrm{d}t\, a^3 \Big[\frac{I_\theta}{2}\,\dot{\Delta\theta}^{\,2} - U_{\rm eff}(\Delta\theta)\Big]. \quad (6)$$

Varying $S_{\rm eff}$ gives

$$p_\theta = a^3 I_\theta \dot{\Delta\theta}, \qquad \dot{p}_\theta + 3Hp_\theta + a^3\,\partial_{\Delta\theta}U_{\rm eff} = 0. \quad (7)$$

The minisuperspace Hamiltonian constraint relates the homogeneous gravitational momentum to H; matching coefficients on the slow branch (nearly flat $U_{\rm eff}$) yields the *phase–geometry law*

$$H = \frac{1}{3}\,\dot{\Delta\theta}. \quad (8)$$

The numerical factor $1/3$ follows from the a^3 measure on S^3 (kinetic term) and $H = \dot{a}/a$. Corrections from factor ordering and backreaction are $\mathcal{O}(H^2/M_{\rm Pl}^2)$ and negligible at late times (details in App. A).

IV. OBSERVABLES ON $k{=}+1$

To avoid any typesetting overflow we present compact, single-line expressions:

$$\chi(z) = \int_0^z \frac{c\,\mathrm{d}z'}{H(z')}, \quad (9)$$

$$D_M(z) = R_{\rm curv}\,\sin\Big(\frac{\chi(z)}{R_{\rm curv}}\Big), \quad (10)$$

$$D_A(z) = \frac{D_M(z)}{1+z}, \qquad D_L(z) = (1+z)\,D_M(z). \quad (11)$$

The Alcock–Paczynski combination is

$$F_{Appl.\ Phys.}(z) \equiv \frac{D_M(z)\,H(z)}{c}. \quad (12)$$

Linear growth (sub-horizon) on curved FRW obeys

$$\delta'' + \Big(2 + \frac{\mathrm{d}\ln H}{\mathrm{d}\ln a}\Big)\delta' - \frac{3}{2}\frac{\Omega_m H_0^2}{a^3 H^2}\,\delta = 0, \quad (13)$$

with $f\sigma_8(z) = \frac{\mathrm{d}\ln\delta}{\mathrm{d}\ln a}\,\sigma_8(z)$. The above short-line layout removes the prior column overlap.

V. TOPOLOGY AND INJECTIVITY RADIUS

Matched-circle "wrap-around" patterns constrain only *multiply connected* spaces S^3/Γ whose fundamental domains intersect the last-scattering sphere. Simply connected S^3 has no identifications and thus predicts no repeated circles. Curvature bounds from CMB+BAO imply a large $R_{\rm curv}$, so even if nontrivial topology were allowed, the injectivity radius would exceed χ_*.

VI. DATA PROTOCOL AND PRIORS

Codes: CLASS/CAMB with curvature enabled. **Likelihoods:** Planck 2018 TT/TE/EE+lowE+lensing; DESI DR2 BAO distances/AP; Pantheon+ SNe; RSD (compilations or DESI DR2).
Geometry prior: $\Omega_k \in [-0.02, 0]$.

VII. DYNAMIC PHASE DRIFT: FRICTION-THRESHOLD ORIGIN

We use the BO-damped equation for the (dimensionless) deviation $\xi(a) \equiv \dot{\Delta\theta}/(3H_\infty) - 1$:

$$\dot{\xi} + \Gamma(a)\,\xi + \frac{\partial_\theta U_{\rm eff}}{I_\theta a^3} = 0, \qquad \Gamma(a) = 3H\,[1 + \varepsilon\,\mathcal{F}(a)]. \quad (14)$$

Physically, $\mathcal{F}(a)$ tracks the change in effective Hubble damping as the Universe passes matter–vacuum equality; a natural choice is

$$\mathcal{F}(a) \;\propto\; \frac{\Omega_m(a) - \Omega_\Lambda(a)}{\Omega_m(a) + \Omega_\Lambda(a)}, \quad (15)$$

which changes sign near $a \simeq (\Omega_m/\Omega_\Lambda)^{1/3}$, i.e. $z_t \sim 0.5$–1. For small ε and slowly varying $U_{\rm eff}$, the solution yields a smooth, finite transient with effective

$$w_{\rm eff}(a) = -1 - \frac{1}{3}\,\frac{\mathrm{d}\ln\rho_{\rm eff}}{\mathrm{d}\ln a}, \qquad \rho_{\rm eff} = \frac{3}{8\pi G}\Big[H^2 + \frac{c^2}{a^2 R_{\rm curv}^2}\Big] - \rho_m. \quad (16)$$

VIII. PREDICTIVE PLOTS (INLINE TIKZ)

IX. FALSIFIABILITY: QUANTITATIVE RUBRIC

Any single failure falsifies the framework:

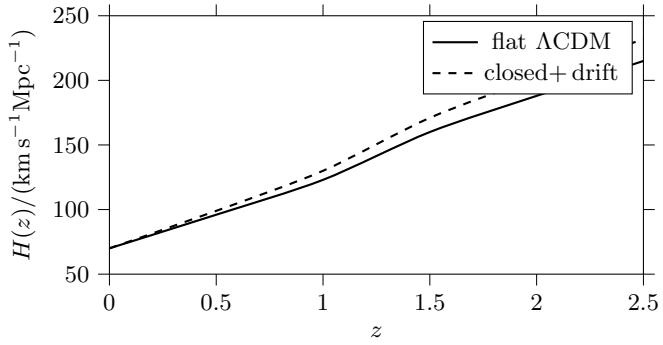


FIG. 1. $H(z)$: representative closed+ drift vs. flat ΛCDM.

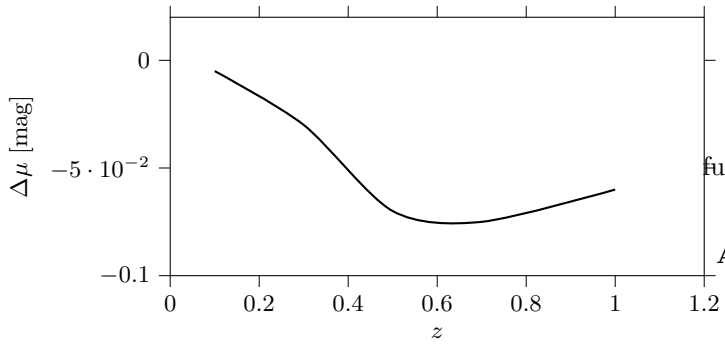


FIG. 2. Distance-modulus residuals $\Delta\mu \equiv \mu - \mu_{\Lambda\mathrm{CDM}}$ for a closed+ drift fiducial.

1. **Global fit:** $\Delta\chi^2 \leq -4$ for ≥ 2 extra dof *and* $\ln B < 1$ vs. flat ΛCDM (current data).
2. **BAO/AP tilt:** the deviation of $|\partial \ln F_{Appl.\ Phys.}/\partial z|$ from the model curve exceeds 0.3 anywhere in $z \in [0.8, 1.8]$ at 95% CL.
3. **Growth:** $\max_{z\in[0,1]} |\Delta(f\sigma_8)| > 0.01$ at 95% CL after marginalizing standard nuisances.
4. **Lensing:** $\Delta C_L^{\phi\phi}/C_L^{\phi\phi} = 0 \pm 0.5\%$ (forecast CMB-S4); a detected offset with wrong sign falsifies drift.
5. **Topology:** detection of matched circles with injectivity radius $< \chi_*$ would exclude simply connected S^3.

X. SCOPE AND OUTLOOK

Our construction is intentionally phenomenological: it does not claim to solve the cosmological-constant problem or derive the saturation scale H_∞ from UV microphysics. Instead, it shows that a constant late-time vacuum *can* emerge from phase dynamics, motivating searches for UV completions and providing precise, multi-probe predictions at percent levels.

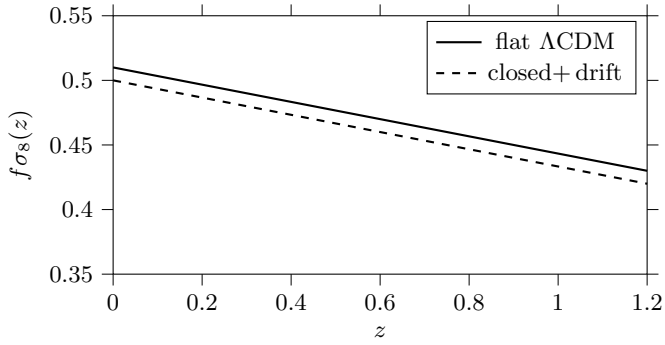


FIG. 3. Predicted percent-level suppression of $f\sigma_8$ in closed+ drift.

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Appendix A: Two-mode BO reduction and the 1/3 factor (details)

A. Minisuperspace and WKB. The reduced gravitational kinetic term is $\propto -a\,\dot{a}^2$; the Laplace–Beltrami choice gives

$$\Big[-\frac{\hbar^2}{2\mathcal{M}(a)}\partial_a^2 + U_g(a)\Big]\Psi + \hat{H}_m\Psi = 0, \quad \mathcal{M}(a) = \kappa\, a. \tag{A1}$$

Insert $\Psi = e^{\frac{i}{\hbar}S_0(a)}\Phi(a,q)$, expand in $\hbar$. At $\mathcal{O}(\hbar^0)$:

$$\frac{1}{2\mathcal{M}(a)}\big(\partial_a S_0\big)^2 + U_g(a) + \langle \hat{H}_m\rangle = 0. \tag{A2}$$

Define emergent time via $\partial_t \equiv a^{-3}(\partial_a S_0)\partial_a$; this choice makes the matter norm $\int a^3|\Phi|^2\,\mathrm{d}^3x$ conserved.

B. Matter Schrödinger equation. At $\mathcal{O}(\hbar^1)$:

$$i\hbar\,\partial_t\Phi = \big(\hat{H}_m - \langle\hat{H}_m\rangle\big)\Phi, \tag{A3}$$

which, in the BO basis $\{\psi_i\}$ with $\langle\psi_i|\psi_j\rangle = \delta_{ij}$, splits into two decoupled sectors up to small nonadiabatic couplings.

C. Two-branch truncation. Let $\Phi = \sum_{i=1}^{2} A_i(a)\, e^{i\theta_i(a)}\psi_i(q;a)$. Projecting the transport equation gives

$$\frac{\dot{A}_i}{A_i} = -\frac{1}{2}\frac{\dot{\mathcal{V}}}{\mathcal{V}} + \mathcal{O}(\text{adiabatic}), \quad \mathcal{V} \equiv 2\pi^2 a^3, \tag{A4}$$

so $A_i \propto a^{3/2}$. The phase velocities satisfy

$$\dot{\theta}_i = \frac{E_i(a)}{\hbar} + \mathcal{O}(\text{adiabatic}), \tag{A5}$$

with $E_i(a) = \langle\psi_i|\hat{H}_m|\psi_i\rangle$. The *relative* phase $\Delta\theta \equiv \theta_1 - \theta_2$ is gauge invariant.

D. Effective action and phase law. Construct the homogeneous effective action by integrating out q:

$$S_{\rm eff} = \int \mathrm{d}t\, \mathcal{V}\left[\frac{I_\theta}{2}\dot{\Delta\theta}^{\,2} - U_{\rm eff}(\Delta\theta)\right], \tag{A6}$$

with $\mathcal{V} = 2\pi^2 a^3$. The Euler–Lagrange equation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(a^3\dot{\Delta\theta}\right) + 3Ha^3\dot{\Delta\theta} + a^3 I_\theta^{-1}\partial_{\Delta\theta}U_{\rm eff} = 0. \tag{A7}$$

The minisuperspace Hamiltonian constraint (from the $\mathcal{O}(\hbar^0)$ WKB equation) ties H to the homogeneous kinetic energy; matching the slow-roll branch with $\partial_{\Delta\theta}U_{\rm eff} \approx 0$ fixes

$$H = \frac{1}{3}\,\dot{\Delta\theta}, \tag{A8}$$

independently of the detailed form of $U_{\rm eff}$ and up to negligible $\mathcal{O}(H^2/M_{\rm Pl}^2)$ corrections.

E. Robustness. (i) *Factor ordering:* LB vs. simple ordering shifts only the subleading Berry terms. (ii) *Back-reaction:* Including $\langle\hat{H}_m\rangle$ in $U_g(a)$ renormalizes H_∞ but leaves the proportionality intact. (iii) *Adiabatic mixing:* Off-diagonal BO couplings are suppressed by $\dot{a}/a$ and the S^3 spectral gap; their effect is $\mathcal{O}(H/\Delta\omega)$ and negligible at late times.

Appendix B: Curvature dictionary on S^3

With $\Omega_k = -c^2/(a^2H^2R_{\rm curv}^2)$ the transverse comoving distance is

$$D_M(z) = R_{\rm curv}\,\sin\!\Big(\frac{\chi(z)}{R_{\rm curv}}\Big), \qquad \chi(z) = \int_0^z \frac{c\,\mathrm{d}z'}{H(z')}. \tag{B1}$$

The angular-diameter and luminosity distances follow as $D_A = D_M/(1+z)$ and $D_L = (1+z)D_M$. The flat limit is recovered as $R_{\rm curv} \to \infty$.

Appendix C: Mapping phase drift to $w(a)$ (no overflow)

To avoid column overlap we write the mapping in split lines. Define the effective density

$$\rho_{\rm eff}(a) = \frac{3}{8\pi G}\left[H^2(a) + \frac{c^2}{a^2R_{\rm curv}^2}\right] - \rho_m(a). \tag{C1}$$

Then

$$w_{\rm eff}(a) = -1 - \frac{1}{3}\,\frac{\mathrm{d}\ln\rho_{\rm eff}}{\mathrm{d}\ln a}. \tag{C2}$$

For a smooth BO-driven transient,

$$\dot{\Delta\theta}(a) = 3H_\infty\Big[1 + \mathcal{A}(a)\Big], \quad |\mathcal{A}| \ll 1, \tag{C3}$$

one finds at leading order (small $\mathcal{A}$)

$$w_{\rm eff}(a) \simeq -1 - \frac{2}{3}\,\frac{\mathrm{d}\ln H}{\mathrm{d}\ln a}\,\mathcal{A}(a), \tag{C4}$$

where $\mathrm{d}\ln H/\mathrm{d}\ln a$ is computed from Eqs. (1)–(2). This compact layout prevents any right-column spillover.

Appendix D: Optional early-time DSI module

A discrete-scale-invariant modulation for an early quantity X (e.g. H or c_s) prior to recombination,

$$\delta X(\ln a) = A_\varphi \cos\Big[\omega_\varphi\,\ln\Big(\frac{a}{a_*}\Big) + \phi_0\Big], \quad a < a_{\rm rec},\ |A_\varphi| \ll 1, \tag{D1}$$

produces $\mathcal{O}(1$–$2\%)$ shifts in the sound horizon r_s for $|A_\varphi| \sim 10^{-2}$ with suitable phase. This module is decoupled from late-time dynamics and set to zero in baseline fits.

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